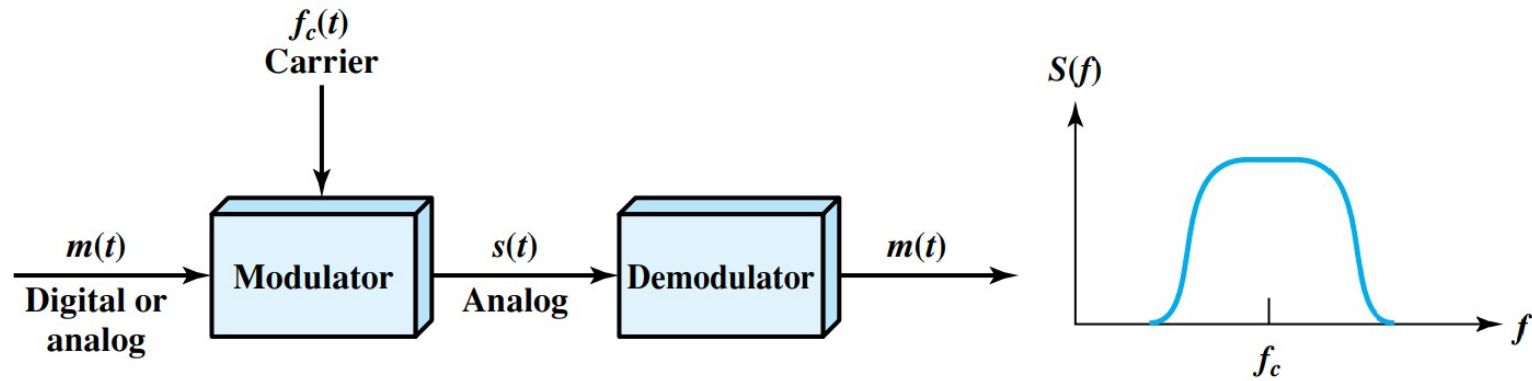


Computer Networks I

Signal Modulation Techniques (Digital to Analog)

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Computer Science and Engineering
IIT Kanpur

Modulation Basics



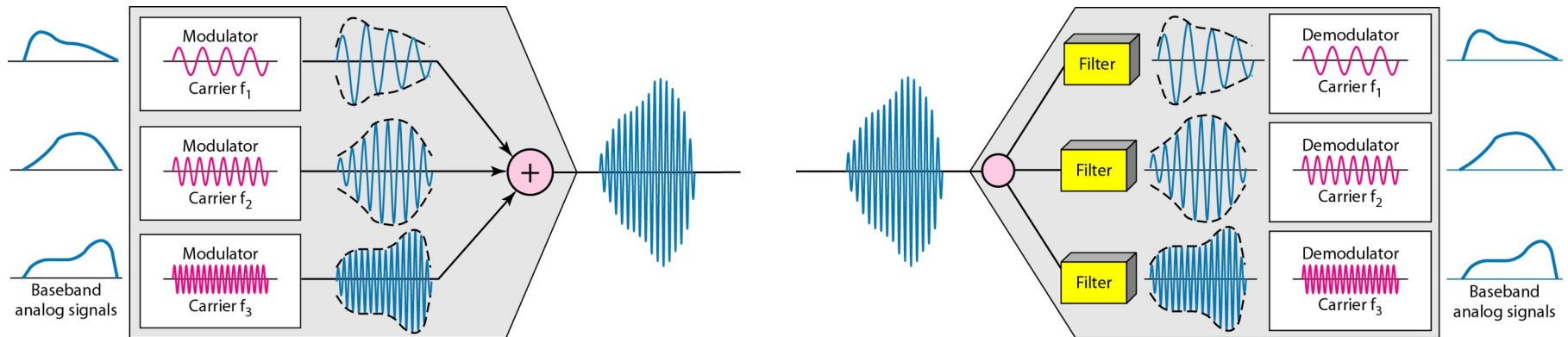
Why Modulation?

□ Frequency reuse and multiplexing

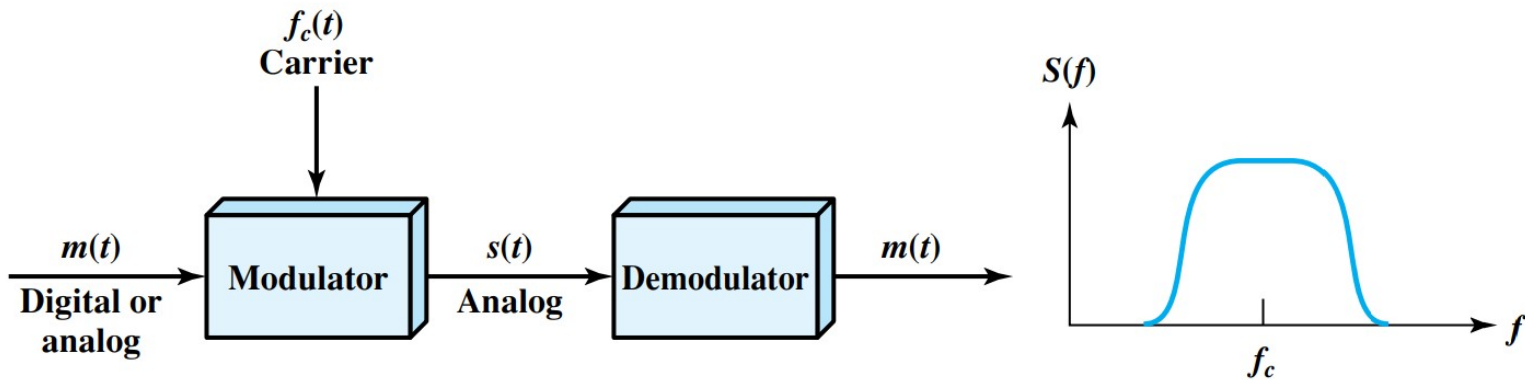
□ Multiple stations can simultaneously transmit at different frequencies

□ Reducing antenna length

□ Antenna length $\propto \lambda$



Digital Data → Analog Signals



- Three principle techniques:

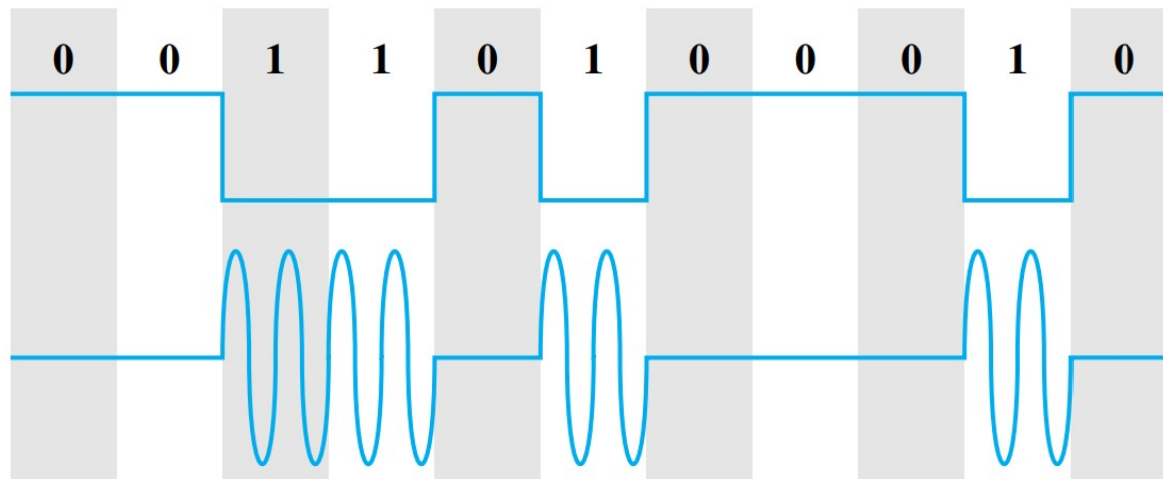
- Amplitude shift keying (ASK)
- Frequency shift keying (FSK)
- Phase shift keying (PSK)

$$s(t) = A \sin(2\pi f t + \phi)$$

Amplitude Shift Keying

- Encode 0/1 by different carrier amplitudes
 - Usually have one amplitude zero

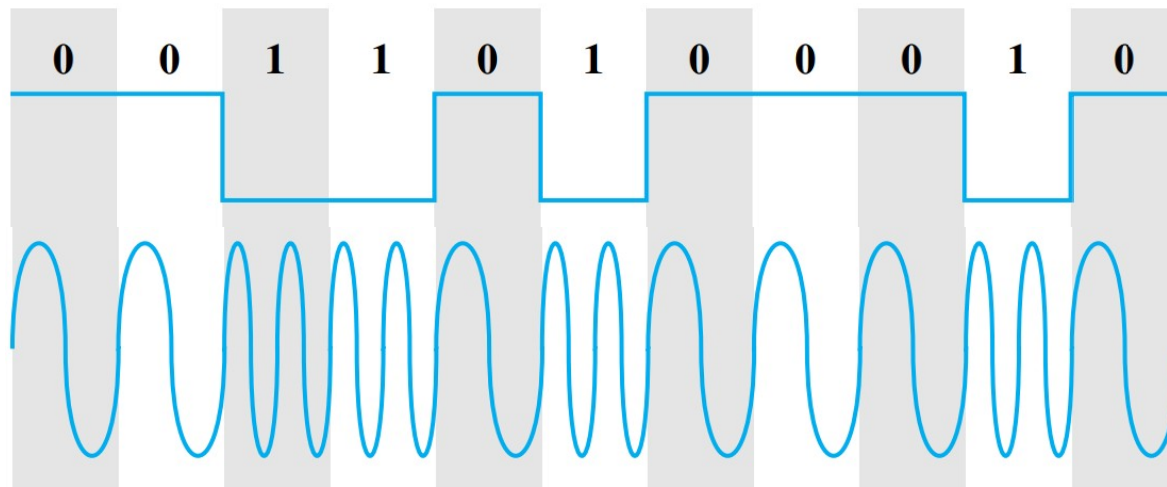
$$s(t) = \begin{cases} A_1 \cos(2\pi f_c t) & \text{binary 1} \\ A_2 \cos(2\pi f_c t) & \text{binary 0} \end{cases} = \begin{cases} A \cos(2\pi f_c t) & \text{binary 1} \\ 0 & \text{binary 0} \end{cases}$$



Frequency Shift Keying

- Most common is binary FSK (**BFSK**)
- Two binary values represented by two different frequencies

$$s(t) = \begin{cases} A \cos(2\pi f_1 t) & \text{binary 1} \\ A \cos(2\pi f_2 t) & \text{binary 0} \end{cases}$$



Frequency Shift Keying

- Most common is binary FSK (**BFSK**)
- Two binary values represented by two different frequencies

$$s(t) = \begin{cases} A \cos(2\pi f_1 t) & \text{binary 1} \\ A \cos(2\pi f_2 t) & \text{binary 0} \end{cases}$$

$$f_i = f_c + (2i - 1 - M)f_d$$

- **MFSK**:
-

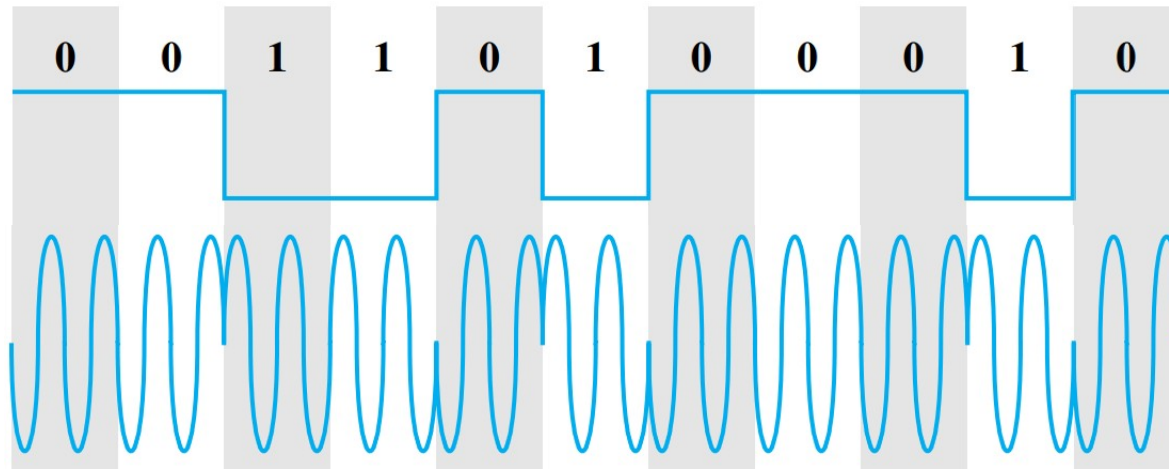
Phase Shift Keying

- Phase of carrier signal is shifted to represent data

$$s(t) = \begin{cases} A \cos(2\pi f_c t + \pi) & \text{binary 1} \\ A \cos(2\pi f_c t) & \text{binary 0} \end{cases}$$

- **Binary PSK**

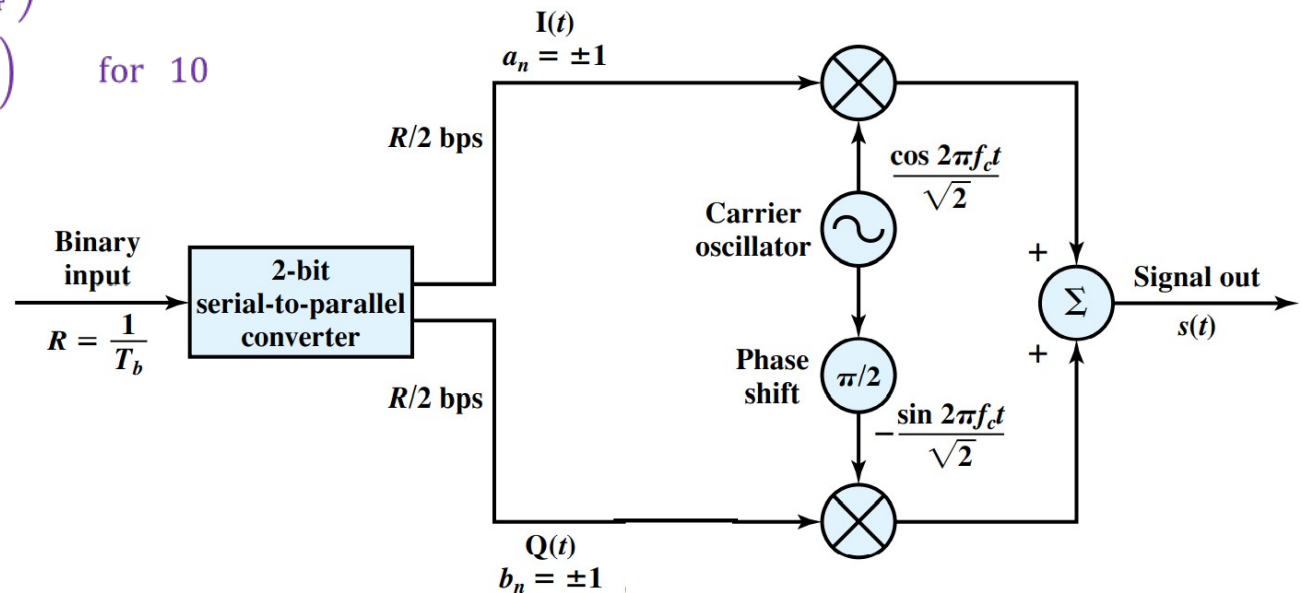
- Two phases represent two binary digits



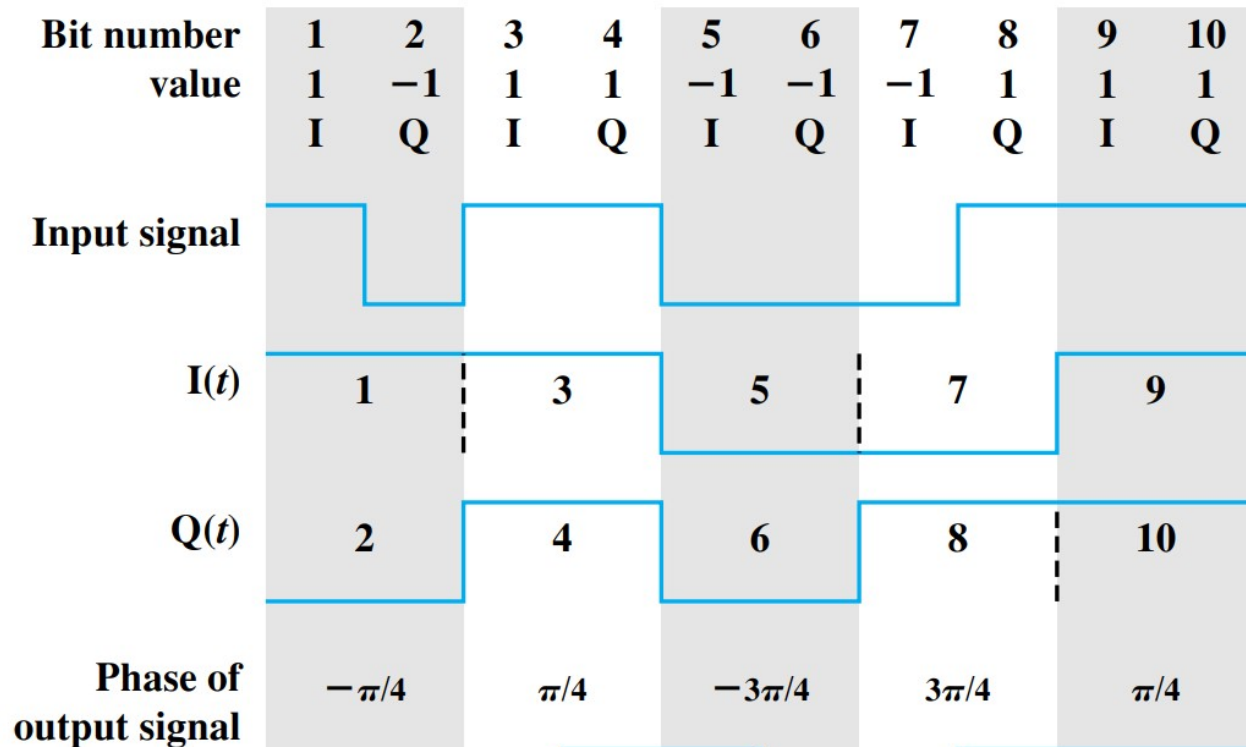
Quadrature Phase Shift Keying

$$s(t) = \begin{cases} A \cos\left(2\pi f_c t + \frac{\pi}{4}\right) & \text{for } 11 \\ A \cos\left(2\pi f_c t + \frac{3\pi}{4}\right) & \text{for } 01 \\ A \cos\left(2\pi f_c t - \frac{3\pi}{4}\right) & \text{for } 00 \\ A \cos\left(2\pi f_c t - \frac{\pi}{4}\right) & \text{for } 10 \end{cases}$$

$$s(t) = \frac{1}{\sqrt{2}} I(t) \cos 2\pi f_c t - \frac{1}{\sqrt{2}} Q(t) \sin 2\pi f_c t$$

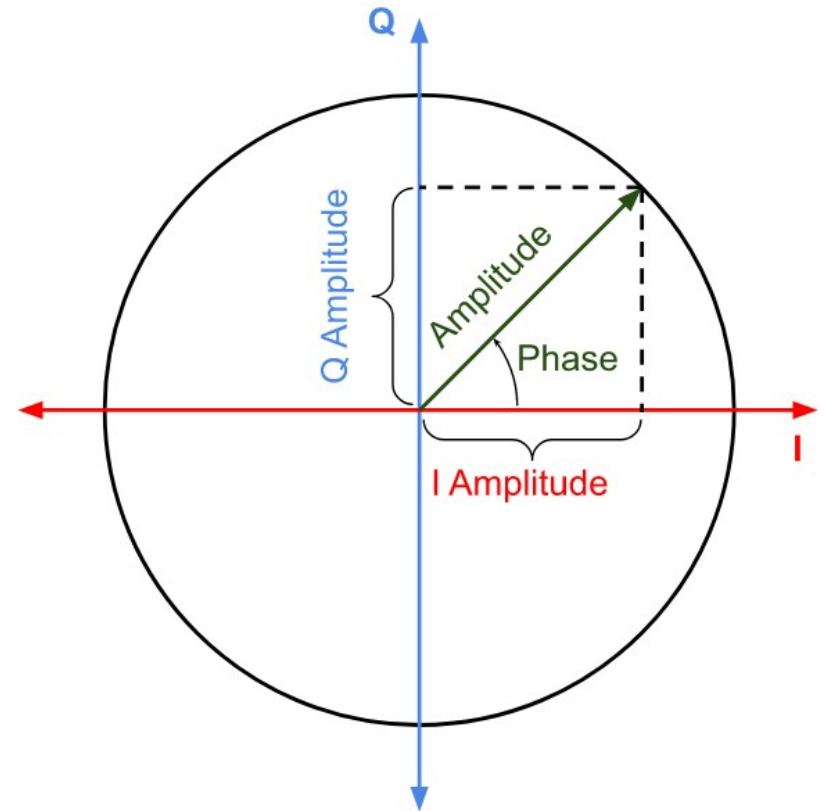


Quadrature Phase Shift Keying



Constellation Diagram

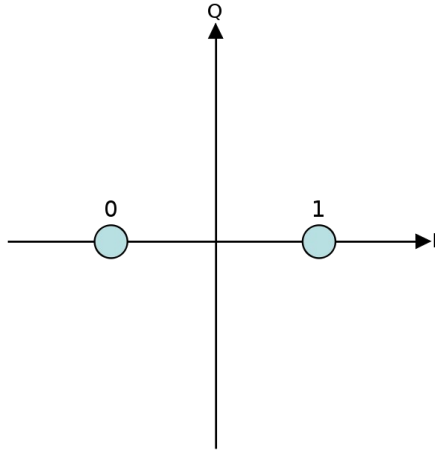
- Representation of a digital modulated signal as a two-dimensional scatter diagram



Src:
https://commons.wikimedia.org/wiki/File:IQ_phasor_diagram.svg

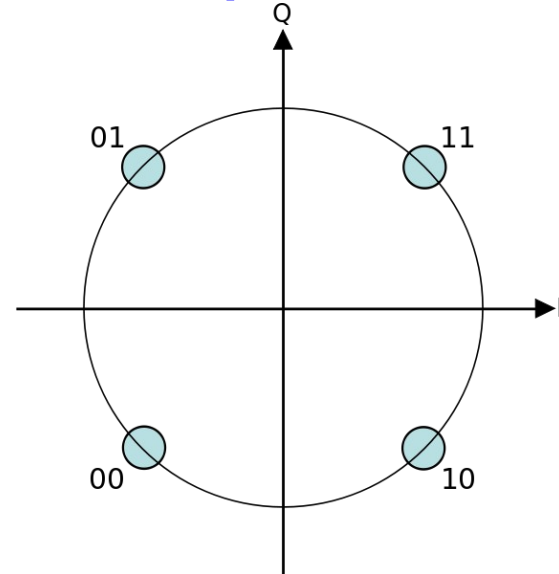
Constellation Diagram

BPSK



Src:https://commons.wikimedia.org/wiki/File:BPSK_Gray_Coded.svg

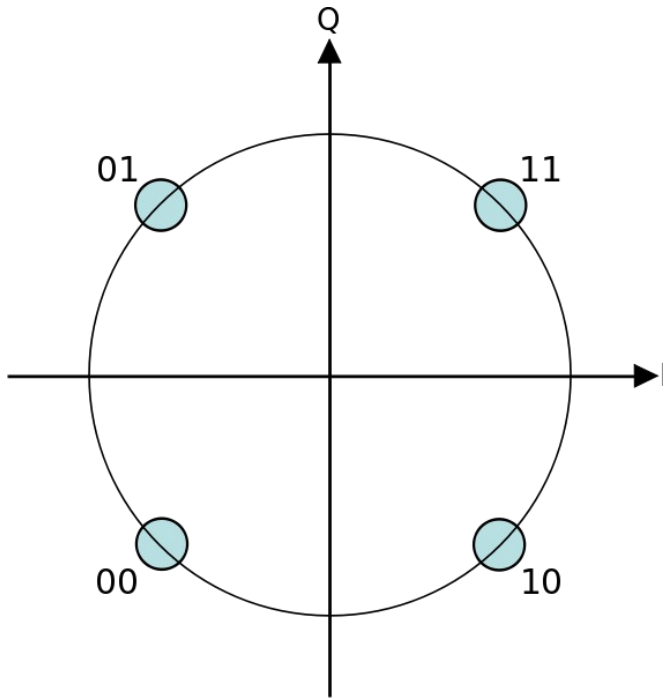
QPSK



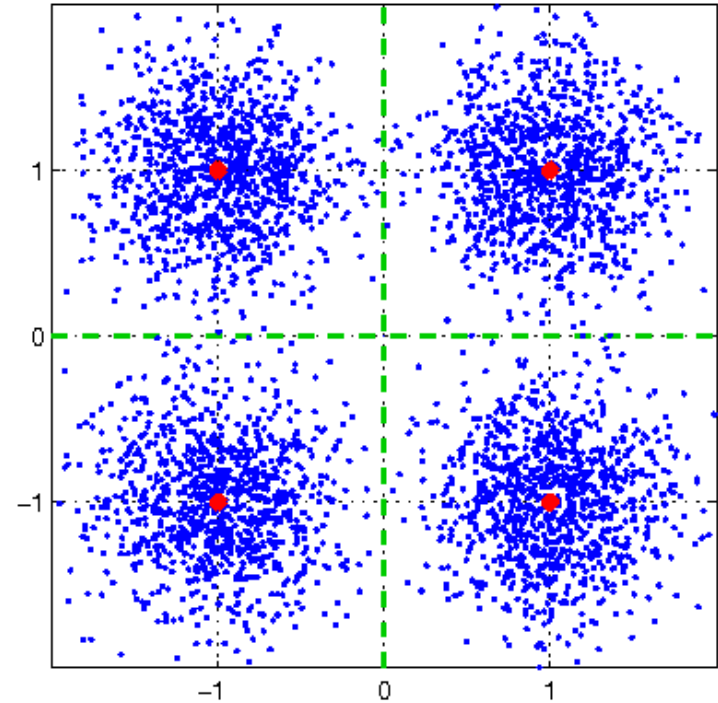
Src:https://commons.wikimedia.org/wiki/File:QPSK_Gray_Coded.svg

Constellation Diagram

QPSK

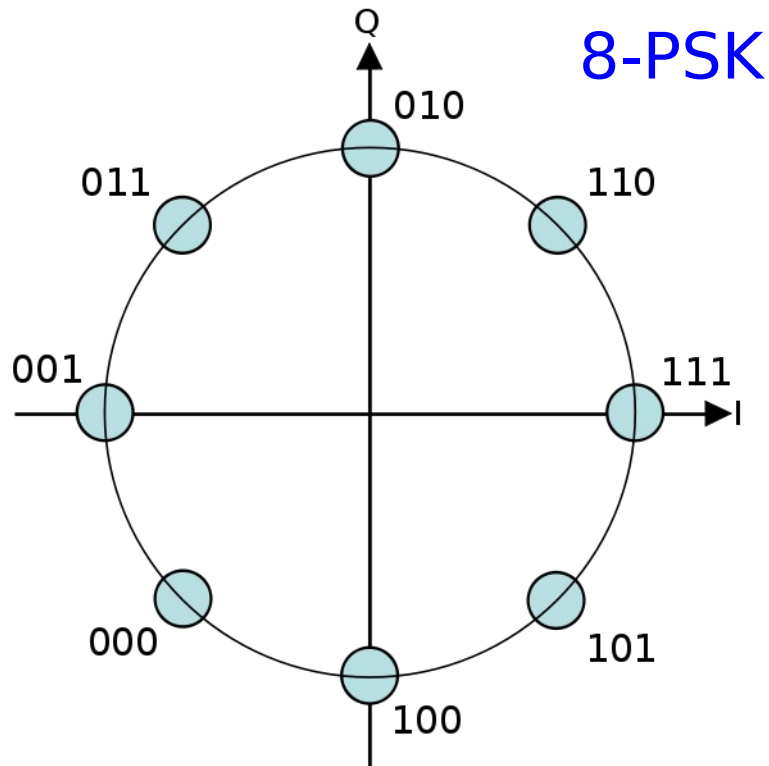


Src:https://commons.wikimedia.org/wiki/File:QPSK_Gray_Coded.svg

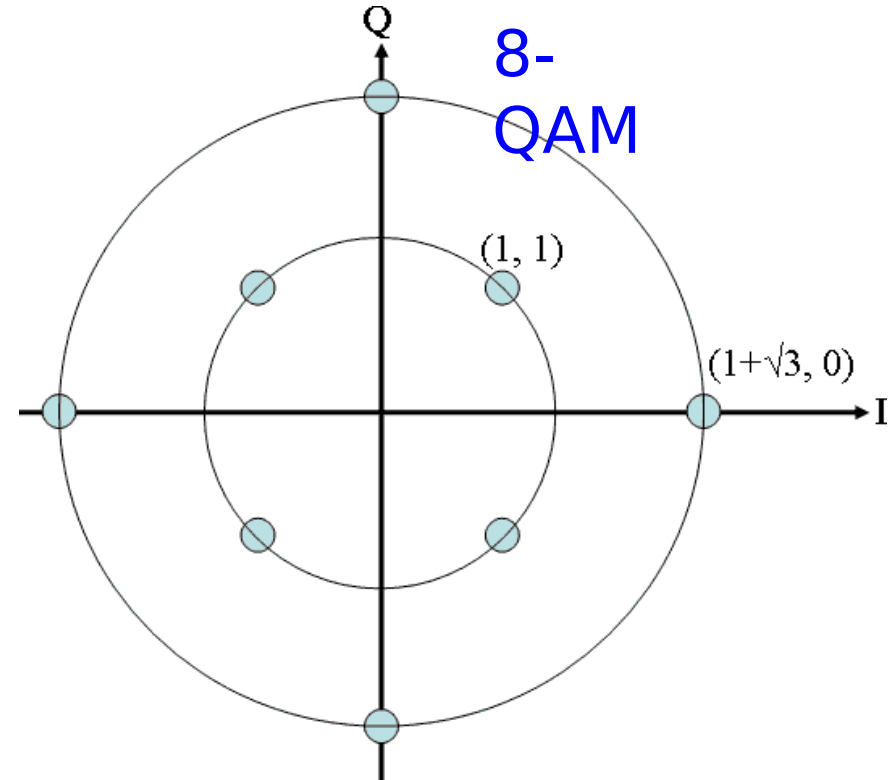


Src:https://commons.wikimedia.org/wiki/File:4qam_constellation_noisy_sigma025.png

Constellation Diagram

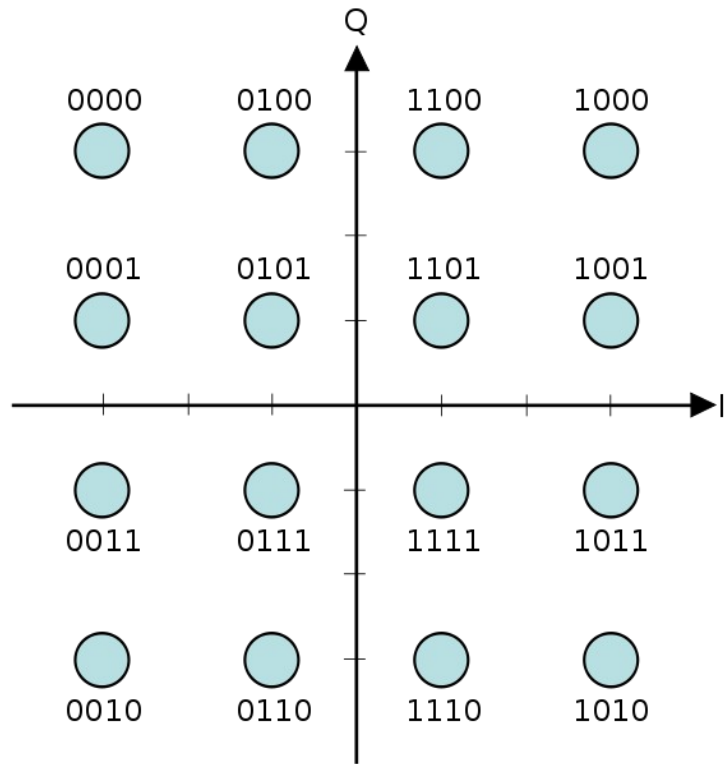


Src:https://commons.wikimedia.org/wiki/File:8PSK_Gray_Coded.svg

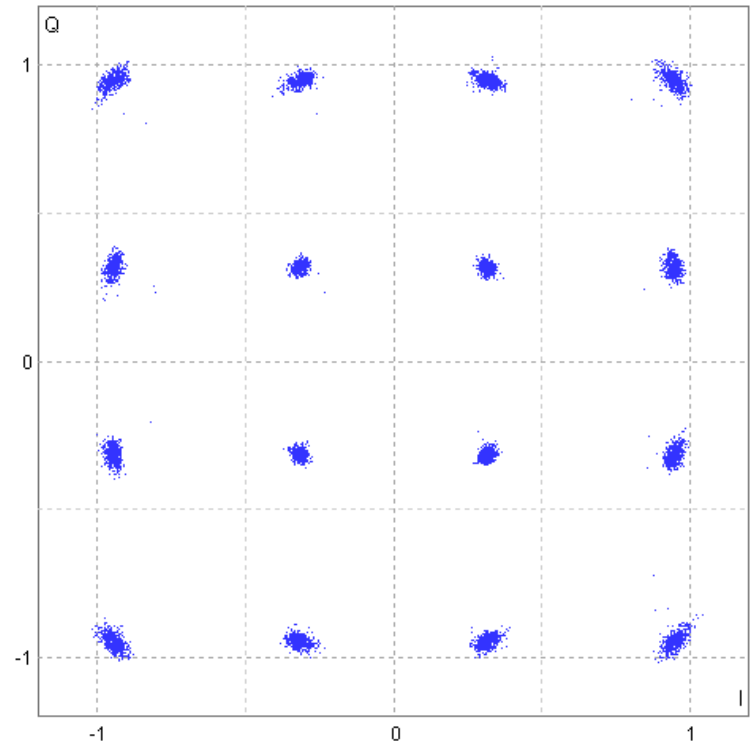


Src:https://commons.wikimedia.org/wiki/File:Circular_8QAM.png

Quadrature Amplitude Modulation



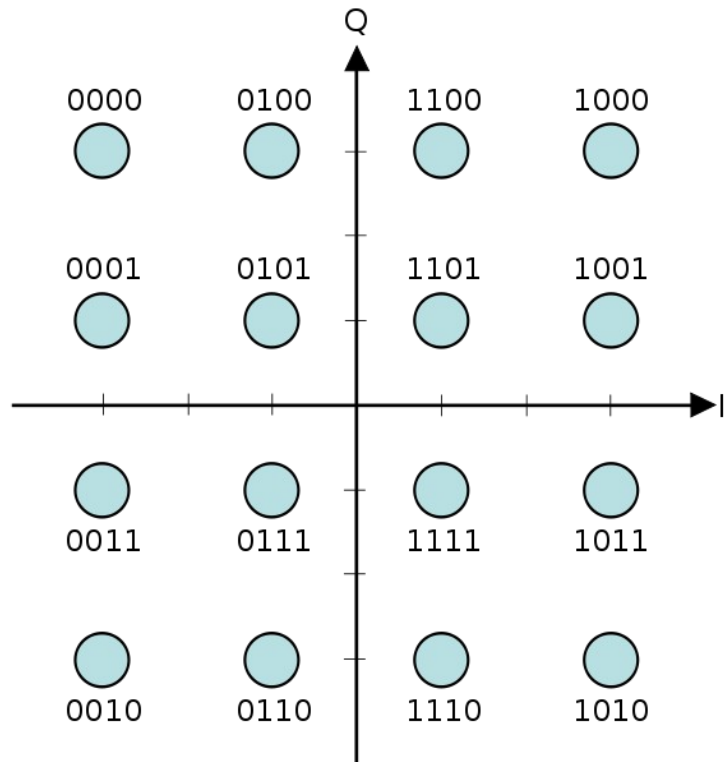
16-QAM



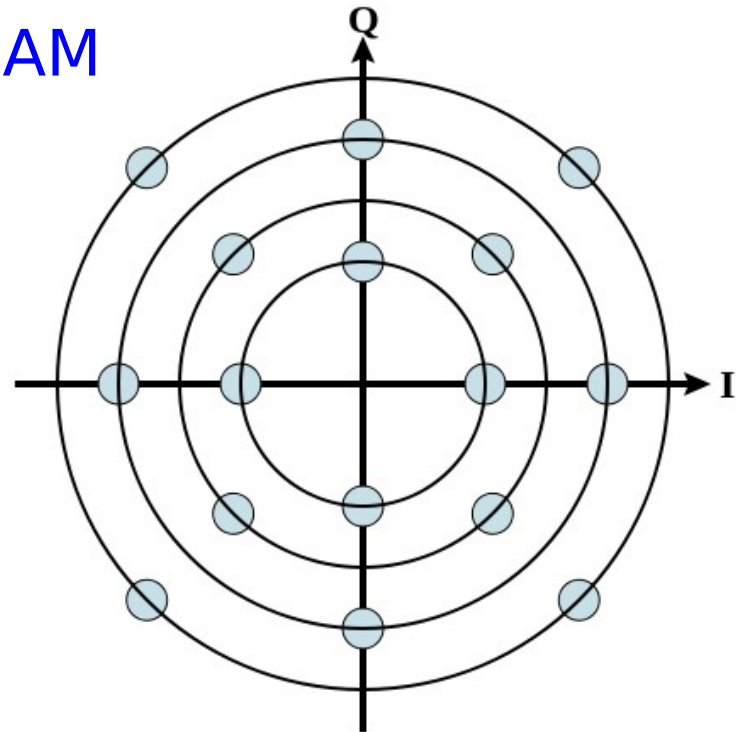
Src:https://commons.wikimedia.org/wiki/File:16QAM_Gray_Coded.svg

Src:https://commons.wikimedia.org/wiki/File:FDD_16-QAM.png

Quadrature Amplitude Modulation



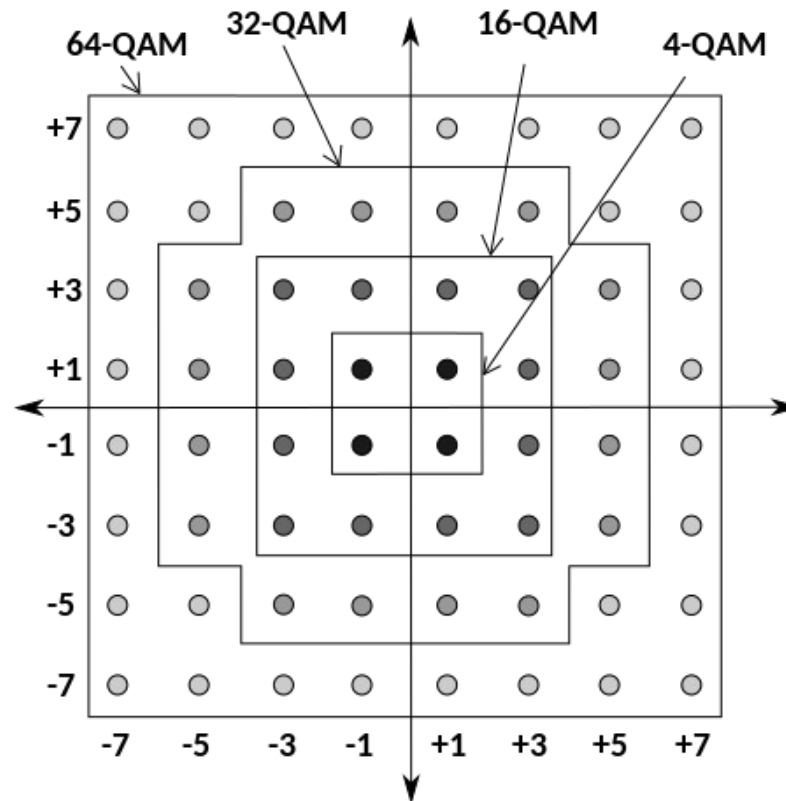
16-QAM



Src:https://commons.wikimedia.org/wiki/File:16QAM_Gray_Coded.svg

Src:https://commons.wikimedia.org/wiki/File:Circular_16QAM.svg

Quadrature Amplitude Modulation



Src: https://commons.wikimedia.org/wiki/File:Rectangular_constellation_for_QAM.svg

Bit and Baud Rate Comparison

Modulation	Bits/Baud	Baud rate	Bit Rate
ASK, FSK, 2-PSK	1	N	N
4-PSK, 4-QAM	2	N	2N
8-PSK, 8-QAM	3	N	3N
16-QAM	4	N	4N
32-QAM	5	N	5N
64-QAM	6	N	6N
128-QAM	7	N	7N
256-QAM	8	N	8N

Summary

□ Modulation techniques (Digital data → Analog signals):

- Different modulation techniques discussed
 - Amplitude Shift Keying
 - Frequency Shift Keying
 - Phase Shift Keying
 - Constellation diagram
 - Quadrature Amplitude Modulation
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